

Workshop

Retrieval

- Given a dataset D of objects, the retrieval task aims to retrieve a subset of objects from D that is most similar to a query q .
- The dataset D could be:
 - Points in R^d
 - Texts, images
 - Complex data objects, e.g., trajectories, time series, or graphs
 - Large Language Model (LLM) embedded database
- To improve retrieval efficiency: Indexing for retrieval using hashing is a technique that uses a mathematical function to map search keys directly to specific storage locations (buckets) for near-instant data access.

Retrieval of Complex Data Objects

- Graph data: Embedding using
 - Learning-based approach GCN [6] & No-learning approach IGK [7] via Euclidean distance
 - Graph Matching Network (GMN) [1] is a powerful learning-based model that generates full-precision embeddings for graph similarity computation.
 - The above are non-hashing baselines
- Trajectory data: no-learning Hausdorff distance and learning-based embedding t2vec [2] as the non-hashing baselines.
- Hashing methods:
 - Voronoi diagram-based VDeH [8],
 - hypersphere-based SpH [3], hyperplane-based brLSH [4], and
 - thresholding-based ITQ [5].

[1] Li, Gu, Dullien, Vinyals, and Kohli. Graph matching networks for learning the similarity of graph structured objects. ICML 2019

[2] Li, Zhao, Cong, Jensen, and Wei. Deep representation learning for trajectory similarity computation. ICDE 2018

[3] Heo, Lee, He, Chang, and Yoon. Spherical hashing: Binary code embedding with hyperspheres. TPAMI 2015

[4] Liu, Zhou, Wu, Zhang, Li, and Li. Refining Codes for Locality Sensitive Hashing. TKDE 2023

[5] Gong, Lazebnik, Gordo, Perronnin. Iterative quantization: A procrustean approach to learning binary codes for large-scale image retrieval. TPAMI 2013

[6] Kipf and Welling. Semi-supervised classification with graph convolutional networks. arXiv:1609.02907 2016

[7] Xu, Ting and Jiang. Isolation graph kernel. AAAI 2019

[8] Xu, Ting, Voronoi Diagram Encoded Hashing. ECML/PKDD, 2025

Discussion

1. Given a vector database, explain how you can improve the retrieval efficiency of a linear search, and what can be expected of the retrieval accuracy.
2. Given a graph database, explain two different approaches to perform the retrieval, and provide examples for each approach. Discuss the pros and cons of the two approaches.
3. Given a trajectory database, explain three different approaches to perform the retrieval, and provide examples for each approach. Discuss the pros and cons of the two approaches.
4. A binary indexing approach often produces a binary hashing code for each vector. Discuss the advantages of this approach over a hashing approach based on real values.

Retrieval Performance Evaluation

- A retrieval method should yield a ranking of nearest neighbors that is highly consistent with the ranking produced by a direct Euclidean distance search on the original dataset or its embedding.
- mAP is the mean of the Average Precision (AP) scores over the set of all queries Q .

The AP for a single query is defined as:

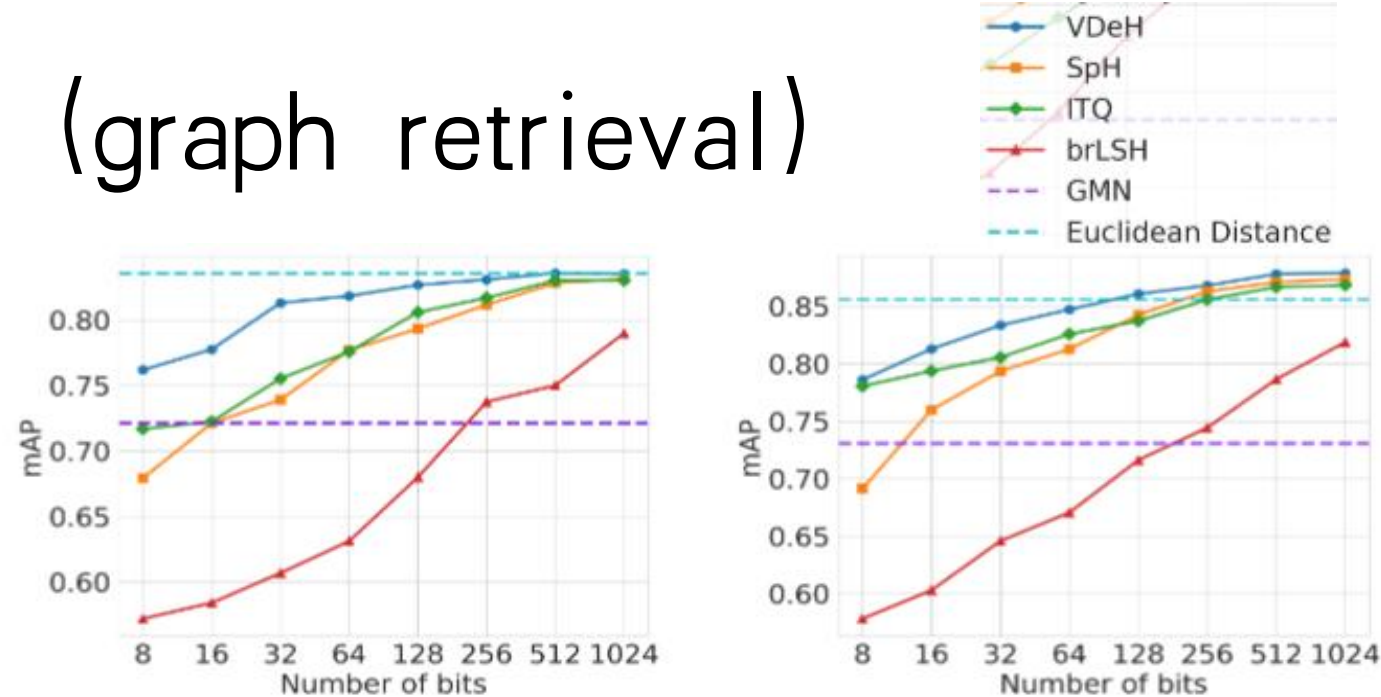
$$AP = \frac{\sum_{k=1}^N (P(k) \times \text{rel}(k))}{\text{Number of relevant items}},$$

where N is the number of items in the database, $P(k)$ is the precision at cut-off k , and $\text{rel}(k)$ is an indicator function defined based on ground truth. The final mAP is then calculated as:

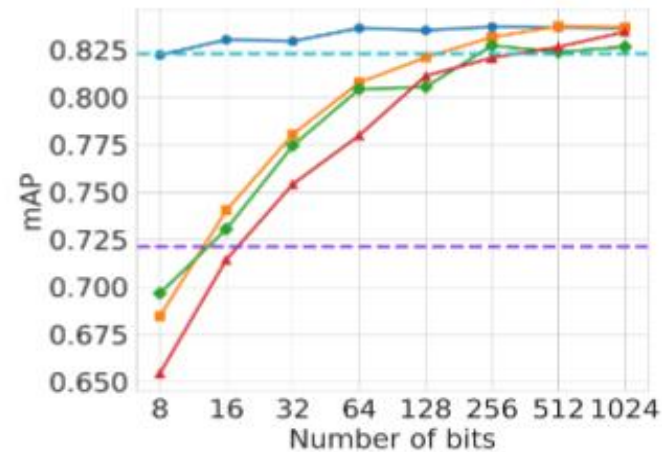
$$\frac{1}{|Q|} \sum_{q \in Q} AP(q).$$

Explain the following (graph retrieval)

1. Compare the baseline results between Euclidean distance and GMN
2. Compare different hashing methods
3. Why can a hashing method (that is an approximation) outperform the performance ceiling achieved by Euclidean distance?

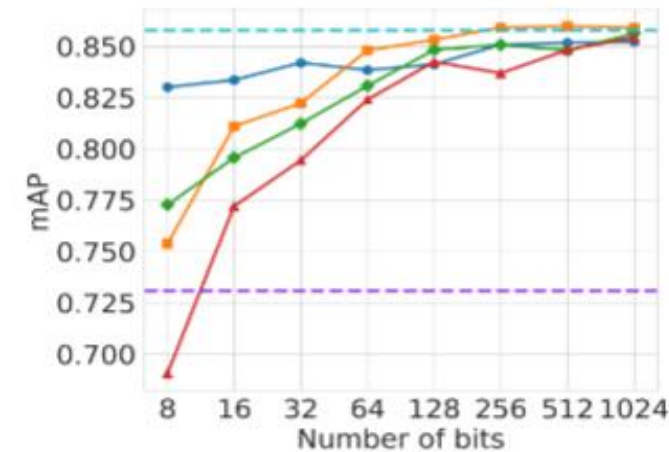


(g) aspirin (IGK)



(j) aspirin (GCN)

(h) salicylic_acid (IGK)



(k) salicylic_acid (GCN)

Retrieval time comparison in trajectory databases

Mapping	Hashing/Distance	T-Drive	Porto	NYC
IDK	VDeH (2048 bits)	33ms	77ms	539ms
	Euclidean Distance	6s ($\sim 182\times$ ↓)	12s ($\sim 156\times$ ↓)	91s ($\sim 169\times$ ↓)
t2vec	Euclidean Distance	6s ($\sim 182\times$ ↓)	13s ($\sim 169\times$ ↓)	92s ($\sim 171\times$ ↓)
None	Hausdorff Distance	41s ($\sim 1242\times$ ↓)	192s ($\sim 2494\times$ ↓)	714s ($\sim 1325\times$ ↓)

Explain the following Anomaly Detection Results

Table 6. Anomaly detection results in terms of AUC on the GeoLife dataset [55].

Anomaly Detection Method	Measure	AUC (%)	Time (s)
<i>k</i> -Nearest Neighbors [30]	IDK+VDeH	<u>99.80</u>	381
	IDK	98.31	4123 ($\sim 10.8\mathbf{x}$ ↓)
	t2vec	70.36	<u>3098</u> ($\sim 8.1\mathbf{x}$ ↓)
	TrajGAT	81.43	6197 ($\sim 16.3\mathbf{x}$ ↓)
	Hausdorff Distance	99.83	535708 ($\sim 1406.1\mathbf{x}$ ↓)
Local Outlier Factor [3]	IDK+VDeH	99.80	473
	IDK	99.98	4605 ($\sim 9.7\mathbf{x}$ ↓)
	t2vec	62.56	<u>3219</u> ($\sim 6.8\mathbf{x}$ ↓)
	TrajGAT	83.72	6482 ($\sim 13.7\mathbf{x}$ ↓)
	Hausdorff Distance	<u>99.89</u>	553958 ($\sim 1171.2\mathbf{x}$ ↓)

What have you learned?

1. Explain the reason why one employs an indexing method such as VDeH.
2. Do any of the methods discussed employ distributional information in the given dataset? If yes, how many levels of distributional information have been utilized?
3. For those methods that have employed only one level of distributional information, how can you employ the second level of distributional information?

Using two levels of distributional information to solve problems involving complex data objects

